SECTION 6 – INTERCONNECT DATA

TABLE OF CONTENTS

6–1 INTRODUCTION

Installation must be in accordance with local and national code.

Section 6, INTERCONNECT DATA, addresses cable interconnections and customer furnished components for the system. It's subsections are broken down as follows:

- 6–1 INTRODUCTION
  - overall system interconnects, component designations
- 6–2 POWER INTERCONNECTS
  - cable connections to the system Main Disconnect Panel and Power Distribution Unit, subsystem power distribution
- 6–3 EMERGENCY OFF WIRING
  - wiring to main disconnect for emergency off
- 6–4 SYSTEM INTERCONNECTS
  - cable interconnects for the system
- 6–5 TSCC ADDITIONAL INTERCONNECTS
  - additional system cable interconnects for each of the TSCC configurations
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- 6–7 CONTRACTOR FURNISHED COMPONENTS
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- 6–8 SPECTROSCOPY INTERCONNECTS
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- 6–9 OXYGEN MONITOR OPTION INTERCONNECTS
  - cable connections for the optional Oxygen Monitor
- 6–10 BRAINWAVE OPTION INTERCONNECTS
  - cable connections for the BrainWave option

Note

Refer to Direction *2261309–100 Advantage Workstation Ultra60 Pre–Installation* for Advantage Workstation information. The BrainWave option offering utilizes an Advantage Workstation.

6–1–1 Component Designators

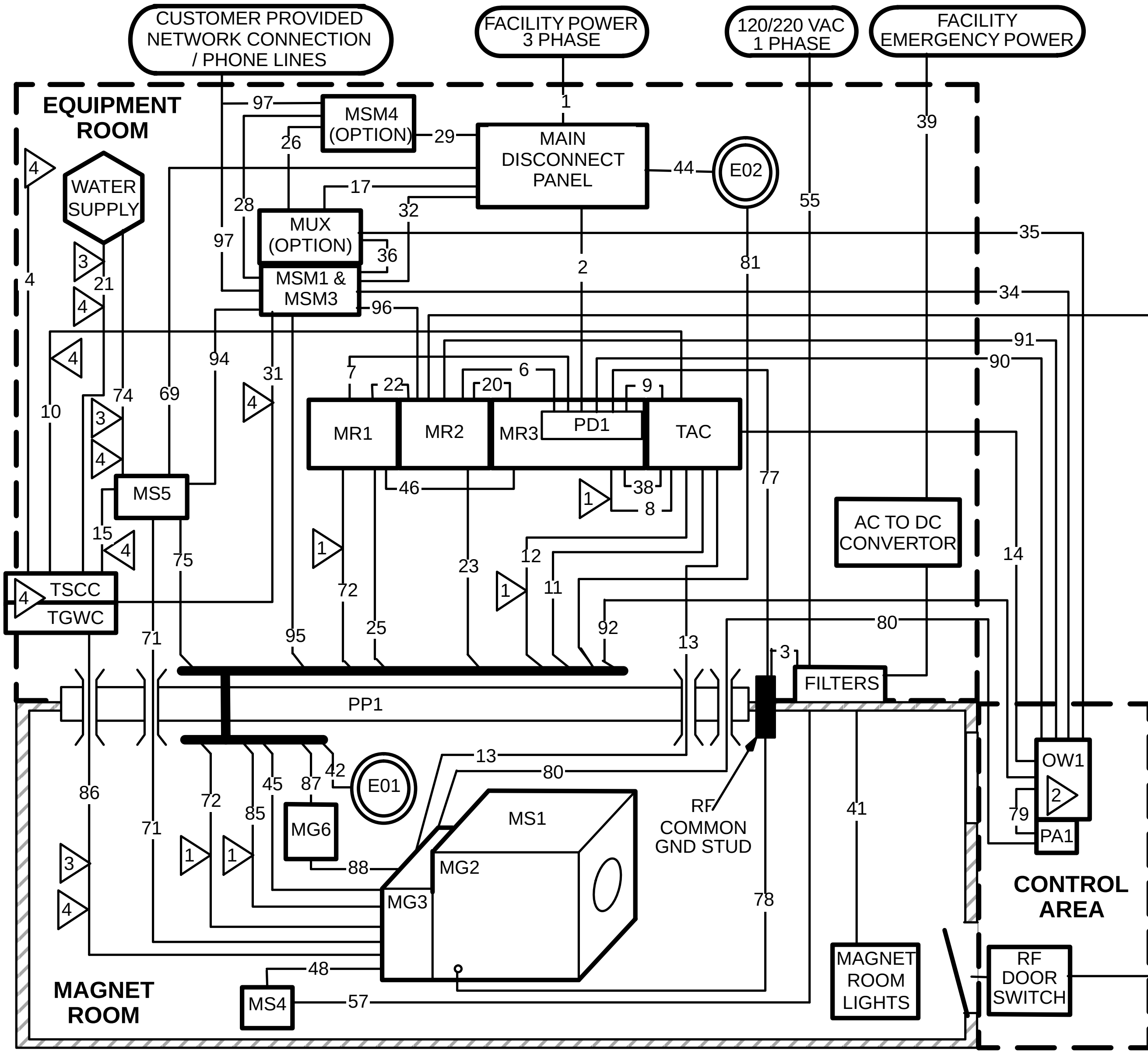
GE uses a Component Designator System as a means of identifying system components in a consistent manner. All subsystem cabinets and other components are referred to by their component designators in the diagrams and tables of this section. For example, the System Cabinet is referred to as MR2. Refer to Table 6–1 for all component designators.

TABLE 6–1
COMPONENT DESIGNATION

BASIC SYSTEM OR OPTION	COMPONENT DESIGNATOR	DESCRIPTION
Basic System	EO1/EO2	Emergency Off Buttons
	MDP	Main Disconnect Panel
	MG2	Magnet Enclosure
	MG3	Magnet Rear Pedestal
	MG6	Blower Box
	MR1	1.5T SRFD II Cabinet
	MR2	System Control Cabinet
	MR3	ACGD/PDU Cabinet has PD1 in lower portion cabinet
	MS1	Superconducting Magnet
	MS4	Magnet Rundown Unit
	MS5	Shield/Cryo Cooler Compressor Cabinet
	MSM1	Magnet Monitor
	MSM3	Modem for Magnet Monitor
	OW1	Operator Workspace
	PD1	Power Distribution Unit (PDU) is a module in lower portion of ACGD/PDU Cabinet
	PA1	Pneumatic Patient Alert Control Box
	PP1	Penetration Panel
	PT1	Patient Transport Table
	RCP	Remote Control Box for Outdoor TSCC, Outdoor TGWC, & Water Cooled TGWC
	TAC	Twin Accessory Cabinet
	TSCC	Twin System Cooling Cabinet (provides water cooling for Shield/Cryo Cooler Compressor & Gradient Coil)
	TGWC	Twin Gradient Water Cooler (provides water cooling for Gradient Coil ONLY; Customer supplied water cooling is required for Shield/Cryo Cooler Compressor Cabinet)
System Options	AW	Advantage Workstation
	AUDIO	BrainWave Audio Console
	BW CABINET	BrainWave Cabinet
	BW CART	BrainWave Cart
	MNS	MNS Cabinet
	MSM4	UPS for Magnet Monitor
	MUX	Phone Line Multiplexer for Magnet Monitor
	OM1	Oxygen Monitor
	OM3	Remote Oxygen Sensor Module

6–1–2 Group Interconnects

Illustration 6–1 shows the Group Interconnect Diagram for 1.5T Signa Twin*Speed* system. Each group contains one or more cables. This diagram should be referred to when using the tables in this section.



1. MUST BE CUT AND CONNECTED TO PP1 AT SITE.
NOTE: IMPEDANCE IS NOT CRITICAL SO EXCESS CABLE SHOULD BE CUT OFF.

2. OPERATOR WORKSPACE (OW1) SUBSYSTEM EQUIPMENT IS PROVIDED WITH MAXIMUM LENGTH CABLES POSSIBLE. SEVERAL OW ASSEMBLIES ARE MOUNTED TO OW TABLE & OW INTERCONNECTS ARE ROUTED THROUGH TABLE CABLE TRAY. FOR REFERENCE USE ONLY THE OW1 RUN NUMBERS ARE 049, 792, 793, 794, 795, 796, 797, 798, 799, 806, & 807.

3. THIS GROUP CONTAINS WATER LINES WHICH SHALL BE ROUTED SEPARATE FROM ELECTRICAL LINES (I.E. POWER & SIGNAL).

4. FOR GROUPS 4, 10, 15, 21, 31, 49, 50, 51, 53, 74, & 86 REFER TO **SECTION 6–5 TSCC ADDITIONAL INTERCONNECTS** OR **SECTION 6–6 TGWC ADDITIONAL INTERCONNECTS** FOR CONFIGURATION APPROPRIATE INTERCONNECTS AND DETAILS.

SIGNA 1.5T TWIN*SPEED* SYSTEM GROUP INTERCONNECT DIAGRAM
ILLUSTRATION 6–1

6–1–3 Definition of Terms

The definition of terms used in Tables 6–2 and 6–3 are:

- Group Number: identifying number referenced to bundles (i.e. groups) of cables as shown in Illustrations 6–1
- Between Units (From/To): component designators as found in Table 6–1
- Area: cross–sectional area of the combined cables in a group

Note

The group area was found by adding up the circular cross–sectional areas of all individual cables within a group. It does not take any fill factors or space between cables into account. Adhere to applicable electrical codes for fill factors.

- Usable Length: total length of a cable MINUS any required take up within cabinets
- Run Number: unique number assigned to each GE–supplied cable

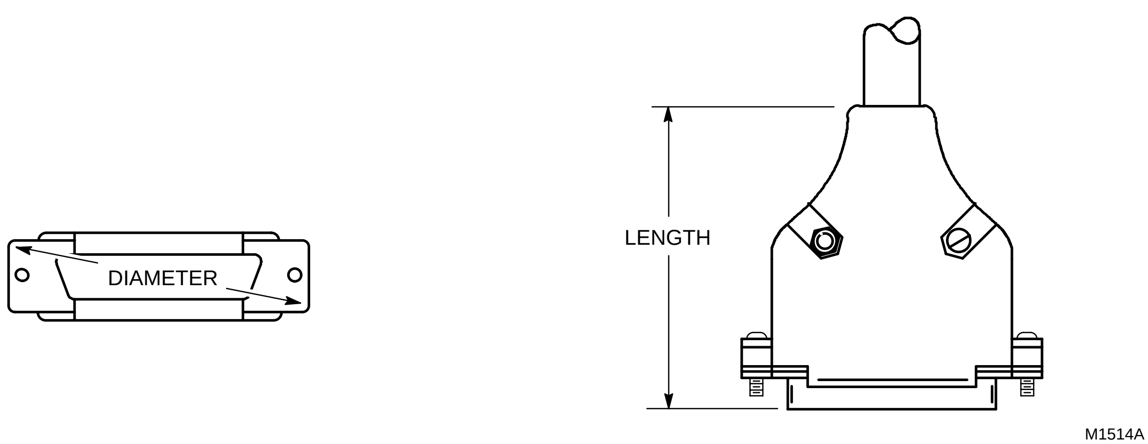
Note

The run number must be used when making special cable orders.

- Cable Diameter: diameter of an individual cable
- Plug Pulling Diameter x Length: cable plug dimensions as shown in Illustration 6–2

Note

In some cases, a cable has more than one connector on an end. These cables will have the number of connectors following the english dimensions of the plug pulling diameter times length (e.g. '2.0x3.25 –x2' means there are 2 connectors with dimensions of 2.0 in. diameter and 3.25 in. length). Of course, the same number of connectors apply to the metric dimensions as well.



FRONT VIEW TOP VIEW

SUBMINIATURE–D CONNECTOR PLUG PULLING DIMENSIONS
ILLUSTRATION 6–2

6–2 POWER INTERCONNECTS

The interconnects for the MDP include:

- Facility Main power; refer to **Section 5 – POWER REQUIREMENTS** for detailed information on main power connections.
- Main power to the system Main Disconnect Panel (MDP) and Power Distribution Unit (PD1).
- Emergency off wiring; refer to **Section 6–3 EMERGENCY OFF WIRING**, for information on the emergency off circuit interfacing.
- System Cooling Cabinet
- Shield/Cryo Cooler Compressor Cabinet (MS5)
- Magnet Monitor equipment (Magnet Monitor, optional UPS for Magnet Monitor, Modem, and optional Multiplexer Box)

Note

The PDU is a module (PD1) in the lower portion of the ACGD/PDU Cabinet (MR3).

The interconnects for the PD1 include:

- Power cables GE–supplied subsystems for Operator Workspace, System Cabinet, Twin Accessory Cabinet, Patient Comfort Compressor.
- 3 auxiliary ground cables to the PD1.
- 2 control cables to the PD1

Conduit or pipe is not recommended for cable runs since the system uses many prefabricated cables with large connectors. However, there may be instances in which conduit is used for power cables. In those cases, cables may be pulled by the lug terminal ends so the connector pulling dimensions on the plug ends will not be a factor for power cables.

Note

The power cables will probably need to be terminated at PD1 located in the lower portion of the Power Cabinet if conduit is used.

Unless otherwise specified, cables and components listed in Table 6–2 are supplied by GE.

6–2 POWER INTERCONNECTS (Continued)

TABLE 6–2
POWER CABLES FROM MDP & PD1

GROUP NUMBER	GROUP AREA in.^{2}(mm^2)	BETWEEN UNITS FROM	BETWEEN UNITS TO	USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm) FROM	PLUG PULLING DIAMETER X LENGTH in. (mm) TO	NOTES
1	—	Facility Power	MDP	—	—	—	—	—	Customer Furnished. **See Note 2.**
2	—	MDP	PD1 See Note 1	—	—	—	—	—	Customer Furnished. Refer to **Section 5–3–2 System Power Distribution Unit** for wire size information.
4	—	MDP/ PDU See Notes	TSCC/ TGWC See Notes	—	—	—	—	—	Refer to **Section 6–5 TSCC ADDITIONAL INTERCONNECTS** or **Section 6–6 TGWC ADDITIONAL INTERCONNECTS** for details & to determine FROM & TO connections.
6	0.803 (518) ◆◆ 0.905 (584) ◆◆	PD1 **See Note 1**	MR2	22 (6.70)	037	0.188 (4.78)	Hard Wired	Hard Wired	#10 AWG / 1 wire Ground cable
					971	0.375 (9.50)	1.60x2.00 (41.1x50.8)	1.60x2.00 (41.1x50.8)	
					972◆◆	0.36 (9.14)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
					968	0.92 (23.37)	Hard Wired	Hard Wired	#8 AWG / 5 wire Power cable
7	1.36 (878)	PD1 **See Note 1**	MR1	22 (6.70)	038	0.188 (4.78)	Hard Wired	Hard Wired	#10 AWG / 1 wire Ground cable
					966	0.92 (23.37)	Hard Wired	Hard Wired	#8 AWG / 5 wire Power cable
					967	0.92 (23.37)	Hard Wired	Hard Wired	#8 AWG / 5 wire Power cable
9	0.43 (278)	PD1 **See Note 1**	TAC	14 (4.27)	989	0.74 (18.80)	Hard Wired	2.33x3.86 (59.2x98.0)	#10 AWG / 4 wire Power cable
17	0.05 (32.3)	MDP	MUX	7 (2.1)	—	0.25 (6.35)	1.00x2.50 (25.4x63.5)	1.00x2.50 (25.4x63.5)	MUX is an option, refer to **Section 2–8–8**

Note 1 The PDU is a module (PD1) in the lower portion of the ACGD/PDU Cabinet (MR3).

2 If low Voltage Step–Up Transformer Option (R4500AW or R4500BE) is used then customer furnished interconnects are required between facility power, transformer and MDP.

◆ This information applies ONLY to Signa 1.5T Twin*Speed* with EXCITE Technology.

◆◆ This information applies ONLY to Signa 1.5T Twin*Speed*.

(Continued)

6–2 POWER INTERCONNECTS (Continued)

TABLE 6–2 (Continued)
POWER CABLES FROM MDP & PD1

GROUP NUMBER	GROUP AREA in.^{2}(mm^2)	BETWEEN UNITS FROM	BETWEEN UNITS TO	USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm) FROM	PLUG PULLING DIAMETER X LENGTH in. (mm) TO	NOTES
29	0.22 (142)	MDP	MSM4	6 (1.83) minus takeup at each end	—	0.375 (9.53)	1.00x2.50 (25.4x63.5)	Hard Wired	UPS Power IN (for Magnet Monitor UPS option)
					939	0.375 (9.53)	1.00x2.50 (25.4x63.5)	1.00x2.50 (25.4x63.5)	UPS Power OUT (for Magnet Monitor UPS option)
32	0.159 (103)	MDP	MSM1 & MSM3	6 (1.8) minus takeup at each end	—	0.375 (9.5)	1.2x3.00 (30.5x76.2)	1.25x2.5 (31.8x63.5)	MSM1 power cable
					—	0.25 (6.35)	2.00x2.50 (50.8x63.5)	2.00x2.50 (50.8x63.5)	Modem for MSM1 power cable. Modem located on top of MSM1. Refer to **Section 2–8–7**.
44	—	MDP	EO2	—	—	—	—	—	Customer Furnished. Refer to **Section 6–3**.
69	1.31 (842)	MDP	MS5	29.8 (9.1) minus takeup at MDP	—	1.29 (32.75)	Hard Wired	Hard Wired	Both ends have liquid tight connector which wires pass through for hard wire connections.
77	0.27 (172)	PD1 **See Note 1**	RF COM-MON GND STUD	—	—	0.584 (14.8)	Ring Terminal	Ring Terminal	Customer supplied ground cable, refer to **Section 5–4–2**.
81	0.460 (297.3)	EO2	PP1	40 (12.2)	296	0.35 (8.9)	Hard Wired	1.30x2.00 (33.5x50.8)	Refer to **Section 6–3**.
90	0.413 (266)	PD1 **See Note 1**	OW1	51 (15.5) **See Note 3**	048	0.188 (4.78)	Hard Wired	Hard Wired	#10 AWG / 1 wire Ground cable
					969	0.70 (17.78)	Hard Wired	Hard Wired	#10 AWG / 4 wire power cable

Note 1 The PDU is a module (PD1) in the lower portion of the ACGD/PDU Cabinet (MR1).

3 M1090ZP, M1090ZN, M3044TC, M3044TA can be ordered to increase usable length of Group 90 to 84 ft (25.6 m).

6–3 EMERGENCY OFF WIRING

6–3–1 Introduction

This section addresses wiring for the Emergency Off circuit (also known as protective disconnect circuit). Refer to **Section 5–3 RECOMMENDED POWER DISTRIBUTION SYSTEM** for information on the recommended protective disconnect device and emergency off button locations and mounting.

The emergency off wiring for the MR system is unique because the wiring into the magnet room must be RF tight.

6–3–2 Main Disconnect Panel Connections

The emergency off circuit is shown in **Section 5–3 POWER DISTRIBUTION SYSTEM** Illustration 5–1. The circuit utilizes the normally closed series loop shown. The MDP provides 2 emergency off buttons.

6–3–3 Magnet Room Wiring

GE provides two cables for routing the emergency off circuit through the Penetration Panel and into the magnet room (Runs 296 and 297). Alternate wiring may be used by the customer; however, the use of these cables ensures that the emergency off wiring will be RF tight.

In Illustration 5–1 black and red wires are used for connections on the ends of Runs 296 and 297. Actually any pair of wires on these runs could be used so long as both ends are consistent with one another. (Runs 296 and 297 are actually nine wire cables.)

6–4 SYSTEM INTERCONNECTS

Table 6–3 contains information on interconnects between all system components. Cables found in Tables 6–2 are not repeated here, although their groups are referenced for completeness. Conduit or pipe is not recommended for cable runs since the system uses many prefabricated cables with large connectors. Unless otherwise specified, cables and components listed in Table 6–3 are supplied by GE.

TABLE 6–3
INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in.^{2}(mm^2)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO				FROM	TO	
1	—	Facility Power	MDP	—	—	—	—	—	Refer to Table 6–2.
2	—	MDP	PD1 **See Note 1**	—	—	—	—	—	Refer to Table 6–2.
3	—	Facility Emerg Power Filter	PP1	—	—	—	—	—	Customer Furnished Ground.
4	—	MDP/ PDU	TSCC/ TGWC	—	—	—	—	—	Refer to **Section 6–5 TSCC ADDITIONAL INTERCONNECTS** OR **Section 6–6 TGWC ADDITIONAL INTERCONNECTS** for details & to determine FROM & TO connections.
5	—	—	—	—	—	—	—	—	Not Used
6	—	PD1	MR2	—	—	—	—	—	Refer to Table 6–2.
7	—	PD1	MR1	—	—	—	—	—	Refer to Table 6–2.
8	3.45 (2221)	MR3	TAC	16 (4.88)	976 977 978	1.21 (30.7) 1.21 (30.7) 1.21 (30.7)	Hard Wired Hard Wired Hard Wired	Hard Wired Hard Wired Hard Wired	Runs 976, 977, & 978 are cut and connected to TAC at site. Total length of 65 ft (19.8 m) is supplied for use in Group 8 & 12 for routing from MR3 to TAC to PP1.
9	—	PD1	TAC	—	—	—	—	—	Refer to Table 6–2.
10	—	TSCC/ TGWC	TAC	—	—	—	—	—	Refer to **Section 6–5 TSCC ADDITIONAL INTERCONNECTS** OR **Section 6–6 TGWC ADDITIONAL INTERCONNECTS** for details & to determine FROM unit.
Note 1 The PDU is a module (PD1) in the lower portion of the ACGD/PDU Cabinet (MR3).									
(Continued)									

6–4 SYSTEM INTERCONNECTS (Continued)

TABLE 6–3 (Continued)
INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in.2(mm2)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO				FROM	TO	
11	0.31 (201)	TAC	PP1	21 (6.40)	611	0.63 (16.0)	1.75x3.80 (44.5x96.5)	1.75x3.80 (44.5x96.5)	Run 611 for system with 2nd Order Resistive Shim Option
12	3.61 (2323)	TAC	PP1	27 (8.2)	979 980 981 988 X 3	1.21 (30.7) 1.21 (30.7) 1.21 (30.7) 0.26 X 3 (6.6 X 3)	Hard Wired Hard Wired Hard Wired Ring Terminal	Ring Terminal Ring Terminal Ring Terminal Ring Terminal	Runs 979, 980, & 981 are cut and connected to TAC at site. Total length of 65 ft (19.8 m) is supplied for use in Group 8 & 12 for routing from MR3 to TAC to PP1. A Ground wire for each of Runs 979, 980, & 981
13	1.27 (819)	TAC	MG2/3	80 (24.4)	—	1.27 (32.3)	Flexible Line	Flexible Line	Run is routed through waveguide in PP1, cut to length at site.
14	0.10 (66)	TAC	OW1	46 (14.0) See Note 3	1004◆or 568◆◆	0.36 (9.14)	1.30x2.00 (33.5x50.8)	2.30x2.10 (58.5x53.5)	Used for 2nd Order Resistive Shim Option
15 **See Note 2**	—	MS5	TSCC	—	—	—	—	—	Refer to **Section 6–5 TSCC ADDITIONAL INTERCONNECTS**
16	—	—	—	—	—	—	—	—	Not Used
17	—	MDP	MUX	—	—	—	—	—	Refer to Table 6–2.
18 to 19	—	—	—	—	—	—	—	—	Not Used

Note 2 This Group contains water lines which shall be routed separate from electrical lines (i.e. power & signal).

3 Run 568 or 1004 can be extended on a 60 Hz system by using M3090RA 60 Hz High Order Shim Cable Extension Kit. This kit includes a 100 ft (30.5 m) extension cable and converters, power supply, etc. A 50 Hz system will need to follow the Special Order Inquiry (SOI) process if extensions are needed.

◆ This information applies ONLY to Signa 1.5T Twin*Speed* with EXCITE Technology.

◆◆ This information applies ONLY to Signa 1.5T Twin*Speed*.

(Continued)

6–4 SYSTEM INTERCONNECTS (Continued)

TABLE 6–3 (Continued)
INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in.^{2}(mm^2)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO				FROM	TO	
20	0.78 (506)	MR2	MR3	14 (4.27)	1034◆ or 710◆◆	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 1034 or 710 is flexible conduit containing fiber optic cable(s) with a minimum bend of 2 in. (50 mm).
21	—	Facility Water Supply	TSCC	—	—	—	—	—	For Indoor Water Cooled TSCC or TGWC ONLY: **Section 6–5–2 Indoor Water Cooled TSCC System Interconnects** or **Section 6–6–2 Indoor Water Cooled TGWC System Interconnects.**
22	0.83 (537)	MR2	MR1	14 (4.27)	229	0.23 (5.84)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	
					702	0.35 (8.90)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
					708◆◆	0.83 (21.1)	0.83x2.00 (21.1x50.8)	0.83x2.00 (21.1x50.8)	Run 708 is flexible conduit containing fiber optic cable(s) with a minimum bend of 2 in. (51 mm).
					774	0.44 (11.2)	2.30x2.00 (58.4x50.8)	2.30x2.00 (58.4x50.8)	
					1033◆	0.83 (21.1)	0.83x2.00 (21.1x50.8)	0.83x2.00 (21.1x50.8)	Run 1033 is flexible conduit containing fiber optic cable(s) with a minimum bend of 2 in. (51 mm).

Note ◆ This information applies ONLY to Signa 1.5T Twin*Speed* with EXCITE Technology.

◆◆ This information applies ONLY to Signa 1.5T Twin*Speed*.

(Continued)

6–4 SYSTEM INTERCONNECTS (Continued)

TABLE 6–3 (Continued)
INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in.2(mm2)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO				FROM	TO	
23	1.54 (997) ◆ 1.06 (686) ◆◆	MR2	PP1	31 (9.45) **See Note 4**	231	0.212 (5.84)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	**RF Receive Cable**
					488	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	**RF Receive Cable**
					489	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	**RF Receive Cable**
					490	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	**RF Receive Cable**
					499	0.234 (5.94)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	**RF Receive Cable**
					711/ 712◆◆	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 711/712 is a flexible conduit containing fiber optic cables with a minimum bend of 2 in. (50 mm).
					1025◆	0.234 (5.94)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	**RF Receive Cable**
					1026◆	0.234 (5.94)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	**RF Receive Cable**
					1027◆	0.234 (5.94)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	**RF Receive Cable**
					1028◆	0.234 (5.94)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	**RF Receive Cable**
					1029◆	0.34 (8.6)	1.6x2.00 (40.6x50.8)	1.6x2.00 (40.6x50.8)	
					1030◆	0.525 (13.3)	2.8x2.00 (71.1x50.8)	2.8x2.00 (71.1x50.8)	
					1045◆	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 1045 is a flexible conduit containing fiber optic cables with a minimum bend of 2 in. (50 mm).

Note 4 M3044TC or M3044TA can be ordered to increase usable length of Group 23 to 56 ft (17.1 m).

◆ This information applies ONLY to Signa 1.5T Twin*Speed* with EXCITE Technology.

◆◆ This information applies ONLY to Signa 1.5T Twin*Speed.*

(Continued)

6–4 SYSTEM INTERCONNECTS (Continued)

TABLE 6–3 (Continued)
INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in.^{2}(mm^2)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO				FROM	TO	
24	—	—	—	—	—	—	—	—	Not Used
25	1.50 (966) ◆	MR1	PP1	21 (6.40)	044	0.20 (5.1)	Ring Terminal	Ring Terminal	Ground cable
	1.76 (1134) ◆◆			**See Note 5**	486◆◆	0.36 (9.1)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
					487◆◆	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
					726◆◆	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
					768	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
					769	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
					770	0.31 (7.9)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
					771	0.415 (10.5)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
					772	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
					773	0.34 (8.64)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
					775	0.212 (5.4)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					776	0.212 (5.4)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					777	0.212 (5.4)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					783	0.64 (16.3)	1.60x2.00 (40.6x50.8)	Ring Terminals	

Note 5 M3044TC or M3044TA can be ordered to increase usable length of Group 25 to 51 ft (15.5 m).

◆ This information applies ONLY to Signa 1.5T Twin*Speed* with EXCITE Technology.

◆◆ This information applies ONLY to Signa 1.5T Twin*Speed*.

(Continued)

6–4 SYSTEM INTERCONNECTS (Continued)

TABLE 6–3 (Continued)
INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in.2(mm2)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO				FROM	TO	
26	—	MSM4	MUX	—	—	—	—	—	ONLY USED WITH UPS FOR MAGNET MONITOR: Customer provided phone line routed through UPS for transient protection.
27	—	—	—	—	—	—	—	—	Not Used
28	0.07 (45.2)	MSM1 & MSM3	MSM4	6 (1.8) **minus** takeup at each end	—	0.30 (7.62)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	ONLY USED WITH UPS FOR MAGNET MONITOR: Customer provided phone line, cable diameter & plug pull information are estimates.
29	—	MSM4	MDP	—	—	—	—	—	Refer to Table 6–2.
30	—	—	—	—	—	—	—	—	Not Used
31	—	MSM1	TSCC/ TGWC	—	—	—	—	—	Refer to **Section 6–5 TSCC ADDITIONAL INTERCONNECTS** OR **Section 6–6 TGWC ADDITIONAL INTERCONNECTS** for details & to determine FROM connection.
32	—	MDP	MSM1 & Modem	—	—	—	—	—	Refer to Table 6–2.
33	—	—	—	—	—	—	—	—	Not Used
34	0.09 (58)	MSM1	OW1	70 (21.3) **minus** takeup at MSM1	942	0.34 (8.64)	0.50x0.75 (12.7x19.1)	0.50x0.75 (12.7x19.1)	
(Continued)									

6–4 SYSTEM INTERCONNECTS (Continued)

TABLE 6–3 (Continued)
INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in.^{2}(mm^2)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO				FROM	TO	
35	—	MUX	OW1 InSite Modem	—	—	—	—	—	Customer provided phone line.
36	0.09 (58)	MSM3	MUX	6 (1.8) **minus** takeup at each end	—	0.34 (8.64)	0.50x0.75 (12.7x19.1)	0.50x0.75 (12.7x19.1)	Modem–MUX phone line
37	—	—	—	—	—	—	—	—	Not Used
38	0.10 (66)	MR3	TAC	14 (4.27)	974	0.36 (9.14)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	Data cable
39	—	Facility Emerg Power	Filter	—	—	—	—	—	Refer to **Section 5–8** for DC Lighting Controller cabling.
40	—	—	—	—	—	—	—	—	Not Used
41	—	Filter	Magnet Room Lights	—	—	—	—	—	Refer to **Sections 5–7** and **5–8**.
42	0.096 (61.94)	PP1	E01	84 (25.6) **minus** takeup at EO1	297	0.35 (8.9)	1.30x2.00 (33.5x50.8)	Hard Wired	Refer to **Section 6–3**.
43	0.096 (61.94)	MR2	RF Door Switch	67 (20.4) **minus** takeup at RF Door Switch **See Note 6**	701	0.35 (8.9)	1.30x2.00 (33.5x50.8)	Hard Wired	RF Door Switch provided by RF Screen Room vendor.
44	—	MDP	EO2	—	—	—	—	—	Refer to Table 6–2.

Note 6 M3044TC or M3044TA can be ordered to increase usable length of Group 43 to 97 ft (29.6 m) **minus** takeup at RF Door Switch.

(Continued)

6–4 SYSTEM INTERCONNECTS (Continued)

TABLE 6–3 (Continued)
INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in.^{2}(mm^2)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO				FROM	TO	
45	2.61 (1681) ◆ 3.02 (1943) ◆◆	PP1	MG2/3	30 (9.15)	262	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					282	0.464 (11.8)	2.30x2.00 (57.2x50.8	2.30x2.00 (57.2x50.8	
					300	0.44 (11.18)	2.30x2.00 (57.2x50.8)	Ring Terminals	
					318	0.35 (8.9)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
					385	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
					387	0.415 (10.5)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
					491	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
					492	0.36 (9.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
					493	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
					494◆◆	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
					495◆◆	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
					612	0.63 (16.0)	1.75x3.80 (44.5x96.5)	1.75x3.80 (44.5x96.5)	Run 612 for system with 2nd Order Resistive Shim Option
					624	0.26 (6.6)	Ring Terminals	1.00x1.38 (25.4x35.1)	
					700◆◆	0.23 (5.94)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					711/ 712	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 711/712 is flexible conduit containing fiber optic cables with a minimum bend radius of 2 in. (51 mm).
					715	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	

Note ◆ This information applies ONLY to Signa 1.5T Twin*Speed* with EXCITE Technology.

◆◆ This information applies ONLY to Signa 1.5T Twin*Speed*.

(Continued)

6–4 SYSTEM INTERCONNECTS (Continued)

TABLE 6–3 (Continued)
INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in.^{2}(mm^2)	BETWEEN UNITS FROM	BETWEEN UNITS TO	USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm) FROM	PLUG PULLING DIAMETER X LENGTH in. (mm) TO	NOTES
45 (contin-ue)	2.61 (1681) ◆	PP1	MG2/3	30 (9.15)	716	0.34 (8.64)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
	3.02 (1943) ◆◆				727◆◆	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
					778	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					779	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					780	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					781	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					782	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					828	0.31 (7.75)	2.30x2.00 (57.2x50.8	2.30x2.00 (57.2x50.8	
					829	0.30 (7.62)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
					1036◆	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
					1037◆	0.23 (5.94)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					1038◆	0.23 (5.94)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					1039◆	0.23 (5.94)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					1040◆	0.23 (5.94)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					1041◆	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
					1042◆	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	

Note ◆ This information applies ONLY to Signa 1.5T Twin*Speed* with EXCITE Technology.

◆◆ This information applies ONLY to Signa 1.5T Twin*Speed*.

(Continued)

6–4 SYSTEM INTERCONNECTS (Continued)

TABLE 6–3 (Continued)
INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in.^{2}(mm^2)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO				FROM	TO	
46	0.541 (349)	MR1	MR3	14 (4.27) **See Note 7**	709	0.83 (21.1)	0.83x2.00 (21.1x50.8)	0.83x2.00 (21.1x50.8)	Run 709 is flexible conduit containing fiber optic cables.
47	—	—	—	—	—	—	—	—	Not Used
48	0.071 (45.6)	MS4	MS1	86 (26.2)	606	0.30 (7.6)	0.65x1.85 (16.5x47.0)	0.65x1.85 (16.5x47.0)	
49	—	Out-door TSCC	See Notes	—	—	—	—	—	Refer to **Section 6–5–1 Outdoor TSCC System Interconnects**
50	—	Out-door TSCC/ TGWC	See Notes	—	—	—	—	—	Refer to **Section 6–5–1 Outdoor TSCC System Interconnects** or **Section 6–6–1 Outdoor TGWC System Interconnects** for details and to determine FROM and TO units.
51	—	Out-door TSCC/ TGWC	RCP	—	—	—	—	—	Refer to **Section 6–5–1 Outdoor TSCC System Interconnects** or **Section 6–6–1 Outdoor TGWC System Interconnects** for details and to determine FROM unit.
52	—	—	—	—	—	—	—	—	Not Used
53	—	Indoor Water Cooled TGWC	RCP	—	—	—	—	—	Refer to **Section 6–6–2 Indoor Water Cooled TGWC**.
54	—	—	—	—	—	—	—	—	Not Used
55	—	Facility Power	Filter	—	—	—	—	—	Customer Furnished Magnet Room power (refer to **Sections 5–1** and **7–4**).
56	—	—	—	—	—	—	—	—	Not Used
57	—	Filter	MS4	—	—	—	—	—	Customer Furnished (refer to **Sections 5–1**).
58 to 68	—	—	—	—	—	—	—	—	Not Used
69	—	MDP	MS5	—	—	—	—	—	Refer to Table 6–2.
70	—	—	—	—	—	—	—	—	Not Used

Note 7 M3044TC or M3044TA can be ordered to increase usable length of Group 46 to 59 ft (18.0 m).

(Continued)

6–4 SYSTEM INTERCONNECTS (Continued)

TABLE 6–3 (Continued)
INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in.2(mm2)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO				FROM	TO	
71	4.28 (2758)	MS5	MS1	38.5 (11.7)	621 622	1.65 (41.9) 1.65 (41.9)	2.00x3.75 (50.8x95.3) 2.00x3.75 (50.8x95.3)	2.00x3.75 (50.8x95.3) 2.00x3.75 (50.8x95.3)	Runs 621 and 622 are continuous helium lines routed through PP1. Bend radius of Runs 621 and 622 is 8 in. (203.2 mm). Cable diameter includes foam insulation installed on lines.
72	1.27 (820)	MR1	MG2/3	68 (20.7) **See Note 8**	235 & 257 236 & 258	0.87 (22.1) 0.59 (15.0)	1.36x2.12 (34.5x53.9) 1.5x2.50 (38.1x63.5)	1.36x2.40 (34.5x61.0) 1.5x2.50 (38.1x63.5)	**These runs are RF Transmit Cables.** Runs 235/257 ans 236/258 must be cut to length and connected via PP1 at site. Usable length provided for routing in Equipment Room and Magnet Room.
73	—	—	—	—	—	—	—	—	Not Used
74	—	Facility Water Supply	See NOTES	—	—	—	—	—	Refer to **Section 6–5–2 Indoor Water Cooled TSCC System Interconnects** or **Section 6–6 TGWC ADDITIONAL INTERCONNECTS** for details and to determine FROM and TO units.
75	0.035 (22)	MS5	PP1	42 (12.8)	623	0.21 (5.28)	1.00x1.38 (25.4x35.1)	Ring Terminals	
76	—	—	—	—	—	—	—	—	Not Used
77	—	PD1 **See Note 1**	RF COM-MON GND STUD	—	– –	—	—	—	Refer to Table 6–2.

Note 1 The PDU is a module (PD1) in the lower portion of the ACGD/PDU Cabinet (MR3).

8 M3044TC or M3044TA can be ordered to increase usable length of Group 72. These catalogs provide the following:
– For Runs 235 & 257: 88 ft (26.8 m) usable length Heliax to be cut to length and connected via PP1 at site for routing in Equipment and Magnet Room. Note this Heliax has a minimum bend radius of 10 in. (254 mm)
– For Runs 236 & 258: One cable with total length 50 ft (15.24 m) and one cable with total length 55 ft (16.76 m) to be used for Runs 236 & 258. Run 236 From MR1 to PP1 requires takeup of 4 ft (1.2 m) at PP1. Run 258 From PP1 to MG2 requires takeup of 4 ft (1.2 m) at PP1 and 4 ft (1.2 m) at MG2.

(Continued)

6–4 SYSTEM INTERCONNECTS (Continued)

TABLE 6–3 (Continued)
INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in.^{2}(mm^2)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO				FROM	TO	
78	0.338 (218)	RF COM-MON GND STUD	MS1 GND STUD	60 (18.29) **minus** takeup at RF Common GND Stud **See Note 9**	040	0.464 (11.79)	Hard Wired	Ring Terminal	1 ground wire.
79	0.013 (8.0)	OW1	PA1	5 (1.5) **minus** takeup at PA1	—	0.13 (3.2)	3.00x3.00 (76.2x76.2)	0.38x1.75 (9.6x44.5)	
80**	0.049 (32.1)	PA1	MG2	97 (29.6) **minus** takeup at PA1	—	0.25 (6.4)	pneumatic tubing	pneumatic tubing	This pneumatic tubing is continuously routed from PA1 through PP1 and MG3 to MG3.
81	—	E02	PP1	—	—		—	—	Refer to Table 6–2.
82 to 84	—	—	—	—	—	—	—	—	Not Used
85	6.899 (4441)	PP1	MG3	40 (12.2)	982 983 984 985 986 987	1.21 (30.7) 1.21 (30.7) 1.21 (30.7) 1.21 (30.7) 1.21 (30.7) 1.21 (30.7)	Ring Terminals Ring Terminals Ring Terminals Ring Terminals Ring Terminals Ring Terminals	Hard Wired Hard Wired Hard Wired Hard Wired Hard Wired Hard Wired	These cables have a minimum bend radius of 8 in. (203 mm). These runs are cut to length and connected at site.

Note ** If installation requires greater than 97 feet (29.6 meters) of pneumatic tubing between the squeeze bulb, located on the front of the Magnet Enclosure, and the Patient Alert Control Box (PA1), located near the Operator's Console or Operator Workspace, an Extender Kit (46–317758P2) must be ordered. The Extender Kit consists of a small Extender Box (to be mounted in Equipment Room) and 95 feet (29.0 meter) of pneumatic tubing.

9 M3044TC or M3044TA can be ordered to increase usable length of Group 78 to 91 ft (27.7 m) **minus** takeup at RF Common GND Stud.

10 M3044TC or M3044TA can be ordered to increase usable length of Group 87 to 71 ft (21.6 m).

(Continued)

6–4 SYSTEM INTERCONNECTS (Continued)

TABLE 6–3 (Continued)
INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in.^{2}(mm^2)	BETWEEN UNITS FROM	BETWEEN UNITS TO	USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm) FROM	PLUG PULLING DIAMETER X LENGTH in. (mm) TO	NOTES
86 **See Note 2**	—	TSCC/ TGWC See Notes	MG2	—	—	—	—	—	Refer to **Section 6–5 TSCC ADDITIONAL INTERCONNECTS** OR **Section 6–6 TGWC ADDITIONAL INTERCONNECTS** for details & to determine FROM connection.
87	0.352 (228)	PP1	MG6	32 (9.76) minus takeup at MG6, **See row Notes & Note 10**	045	0.195 (4.95)	Ring Terminal	Ring Terminal	The usable length is based on MG6 being mounted on floor and floor routing of cables.
					784	0.64 (16.3)	Ring Terminals	2.28x3.85 (57.9x97.8)	
88	15.9 (8544)	MG6	MG3	13 (3.95) minus takeup	—	4.5 (104.3)	Flexible vinyl hose	Flexible vinyl hose	The flexible vinyl hoses are cut to length during installation.
89	—	—	—	—	—	—	—	—	Not Used
90	—	PD1	OW1	—	—	—	—	—	Refer to Table 6–2.
91	0.55 (355) ◆ 0.54 (302) ◆◆	MR2	OW1	51 (15.5) **See Note 11**	789	0.44 (11.2)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
					791◆◆	0.44 (11.2)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
					818◆◆	0.44 (11.2)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	OW1 end of cable has 3 connectors, each connector plug pull dimensions are 2.3x2.0 (57.2x50.8).
					836◆◆	0.34 (8.64)	0.50x0.75 (12.7x19.1)	0.50x0.75 (12.7x19.1)	Run 836 is a fiber optic cable with a minimum bend of 4 .7 in. (120 mm).
					1031◆	0.34 (8.64)	0.50x0.75 (12.7x19.1)	0.50x0.75 (12.7x19.1)	
					1032◆	0.525 (13.3)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
					1044◆	0.34 (8.64)	0.50x0.75 (12.7x19.1)	0.50x0.75 (12.7x19.1)	

Note ◆ This information applies ONLY to Signa 1.5T Twin*Speed* with EXCITE Technology.

◆◆ This information applies ONLY to Signa 1.5T Twin*Speed*.

2 This Group contains water lines which shall be routed separate from electrical lines (i.e. power & signal).

11 M1090ZP, M1090ZN, M3044TC, M3044TA can be ordered to increase usable length of Group 91 to 81 ft (24.7 m).

(Continued)

6–4 SYSTEM INTERCONNECTS (Continued)

TABLE 6–3 (Continued)
INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in.^{2}(mm^2)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO				FROM	TO	
92	0.223 (144) ◆ 0.152 (99) ◆◆	PP1	OW1	53 (16.1) **See Note 12**	788 838◆	0.44 (11.2) 0.3 (7.6)	2.30x2.00 (57.2x50.8) 1.30x2.00 (33.0x50.8)	2.30x2.00 (57.2x50.8) 1.30x2.00 (33.0x50.8)	
93	—	—	—	—	—	—	—	—	Not Used
94	0.09 (58.6)	MSM1	MS5	54 (16.4) minus takeup at MSM	826	0.34 (8.64)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
95	0.22 (144)	MSM1	PP1	54 (16.4) **minus** takeup at MSM1	824 825	0.44 (11.2) 0.30 (7.62)	2.30x2.00 (57.2x50.8) 1.30x2.00 (33.5x50.8)	2.30x2.00 (57.2x50.8) 1.30x2.00 (33.5x50.8)	
96	0.09 (58.6)	MSM1	MR2	54 (16.4) **minus** takeup at MSM1	823	0.34 (8.64)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
97	—	Phone Line Con-nection	MSM1 or MSM4 (Op-tion)	—	—	—	—	–	Refer to Section **Section 2–8–8** for additional customer network and/or phone line information. WITH UPS FOR MAGNET MONITOR OPTION: Customer provided phone line routed through UPS for transient protection, refer to Group 26.
Note 12 M1090ZP, M1090ZN, M3044TC, M3044TA can be ordered to increase usable length of Group 92 to 88 ft (26.8 m).									

6–5 TSCC ADDITIONAL INTERCONNECTS

The following sections show the system interconnects required for the indicated Twin System Cooling Cabinet (TSCC) configurations in addition to the interconnects in **Section 6–4 SYSTEM INTERCONNECTS**.

6–5–1 Outdoor TSCC System Interconnects

Note

Refer to **Section 2–7–1 TSCC Configurations Siting Considerations** Table 2–5 for listing of responsibility for the specific installation tasks. Also refer to *Ellis and Watts Technical Publication 470 Pre installation Installation Operating LC 20MT/O Chilled Water System for GE Signa TwinSpeed MRI Equipment* for additional information and details.

The site design for the Outdoor TSCC installation must meet the following requirements for vertical separation and water lines/hose lengths limitations. **Installation of the Outdoor TSCC must be in accordance with local and national codes**.

Vertical Separation Requirements

- Outdoor TSCC must not exceed 100 ft (30.5 m) vertical separation from the Shield/Cryo Cooler Compressor Cabinet **and** the Gradient Coil.

Shield/Cryo Cooler Compressor Water Cooling Lines & Hoses Requirements

Customer must provide water supply and return lines between the Outdoor TSCC and the GE supplied hoses for the Shield/Cryo Cooler Compressor.

- Ellis And Watts provides two 0.75 in. (19 mm) FNPT fittings at the Outdoor TSCC for the Shield/Cryo Cooler Compressor water cooling lines connections, refer to Illustration 2–14 for location of fittings on TSCC. The Outdoor TSCC has ball valves located inside the cabinet. Customer to provide 0.75 in. (19 mm) MNPT connections and 0.75 in. (19 mm) ID copper lines.
- Ellis And Watts provides two 0.5 inch (12.7 mm) ID rubber hoses of 60 ft (18.3 m) total length [usable length of 55 ft (16.7 m) minus takeup for ball valve and hose barb location] to connect from Customer provided copper line hose barbs with ball valves to Shield/Cryo Cooler Compressor Cabinet. Site layout for the Shield/Cryo Cooler Compressor water cooling lines and hoses must meet the following:
  - Total line length from the Outdoor TSCC connection to the Shield/Cryo Cooler Compressor connection MUST NOT EXCEED 160 ft (48.8 m). Ellis And Watts supplied rubber hoses for Shield/Cryo Cooler Compressor is a maximum length of 60 ft (18.3 m) total length [55 ft (16.8 m) usable length].
  - If the 0.75 in. (19 mm) copper lines exceed 100 ft (30.5 m) then the rubber hose length must be reduced 1 ft (0.3 m) for every 1 ft (0.3 m) of copper line that exceeds 100 ft (30.5 m).
  - Copper lines must be insulated outdoors.

6–5–1 Outdoor TSCC System Interconnects (Continued)

Gradient Coil Water Cooling Lines & Hoses Requirements

- Ellis And Watts provides two 1.0 in. (25.4 mm) FNPT fittings at Outdoor TSCC for the Gradient Coil water cooling lines connections, refer to Illustration 2–14 for location of fittings on TSCC. The Outdoor TSCC has ball valves located inside the cabinet. Customer to provide 1.0 (25.4 mm) MNPT connections and 1.0 in. (25.4 mm) ID copper lines.
- GE Medical Systems provides two 3/4 inch (19 mm) ID rubber hoses of 100 ft (30.5 m) total length [usable length of 80 ft (24.4 m) minus takeup for ball valve and hose barb location] to connect from Customer provided copper line hose barbs with ball valves to Gradient Coil inside the Magnet Enclosure. Site layout for the Gradient Coil water cooling lines and hoses must meet the following:
  - Total line length from the Outdoor TSCC to the Gradient Coil MUST NOT EXCEED 200 ft (61 m) from TSCC connection to Gradient Coil connection. GE supplied rubber hoses for Gradient Coil is a maximum length of 100 ft (30.5 m) total length [80 ft (24.4 m) usable length].
  - If the 1 in. (25.4 mm) copper lines exceed 100 ft (30.5 m) then the rubber hose length must be reduced 1 ft (0.3 m) for every 1 ft (0.3 m) of copper line that exceeds 100 ft (30.5 m).

RCP Data Cables Requirements

- Ellis & Watts provides three data cables that connect between the RCP and the Outdoor TSCC. These cables are 100 ft (30.5 m) total length. Usable length is dependent on TSCC and RCP placement [total length minus height of connection at Outdoor TSCC and height of connection at RCP, no cable takeup inside of TSCC or RCP].

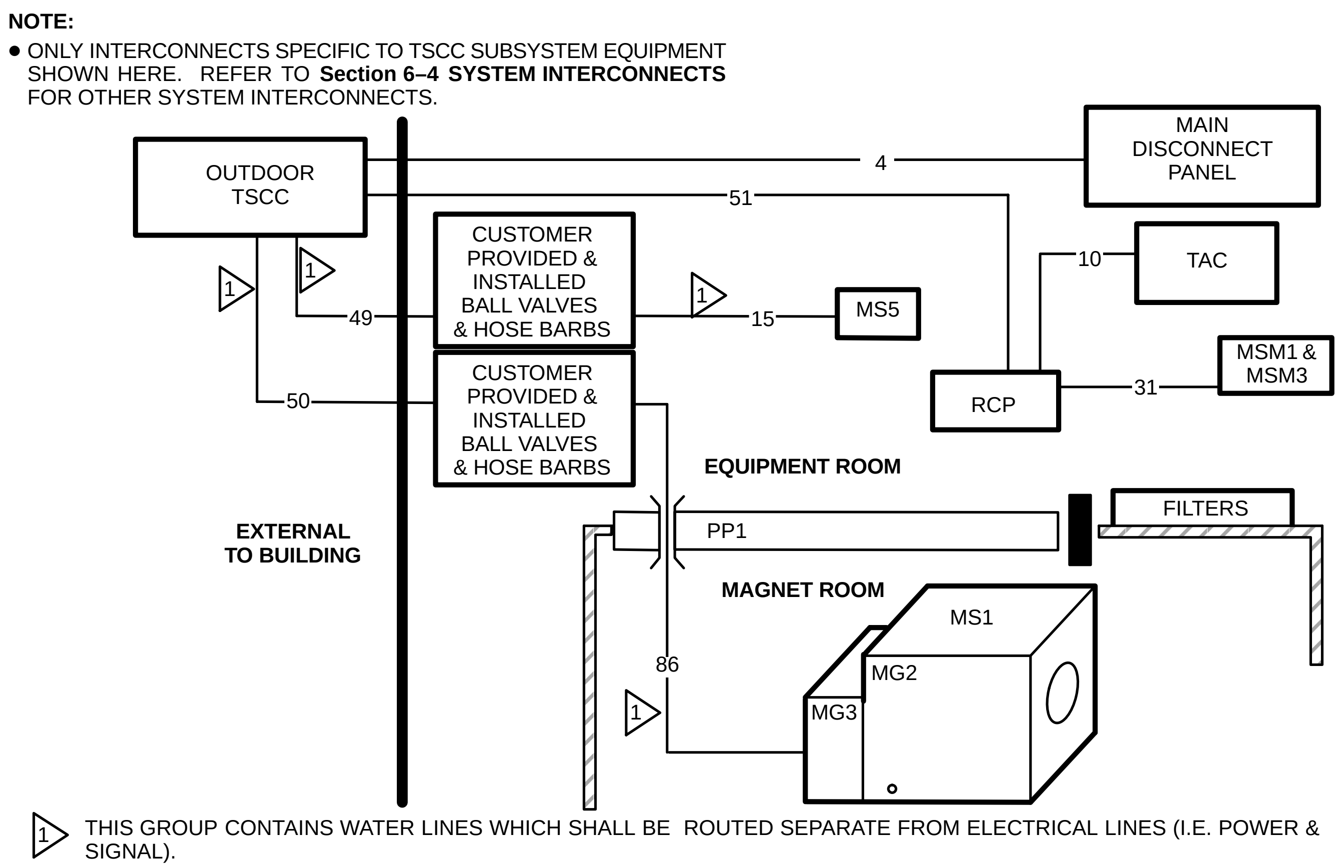
Note

Contact Ellis & Watts International to determine if additional length cable are possible based on specific site design.

System Additional Interconnects

Illustration 6–3 shows the Outdoor TSCC subsystem additional system Group Interconnect Diagram. Each group contains one or more cables, refer to Table 6–4 for specifics.

6–5–1 Outdoor TSCC System Interconnects (Continued)



OUTDOOR TSCC & RCP SUBSYSTEM GROUP INTERCONNECT DIAGRAM
ILLUSTRATION 6–3

6–5–1 Outdoor TSCC System Interconnects (Continued)

TABLE 6–4
OUTDOOR TSCC ADDITIONAL SYSTEM INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in.^{2}(mm^2)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO				FROM	TO	
4	—	MDP	Outdoor TSCC	—	—	—	—	—	Power wiring Customer Furnished.
10	0.11 (71)	RCP	TAC	50 (15.24)	992	0.375 (9.50)	1.60x2.00 (41.1x50.8)	1.60x2.00 (41.1x50.8)	Data cable
15 SEE NOTE 1	1.666 (1078)	Customer provided ball valves & hose barbs for Shield/Cryo Cooler	MS5	55 (16.7) minus takeup at hose barbs See Notes	— —	0.75 (19.0) 0.75 (19.0)	Flexible Tubing Flexible Tubing	Flexible Tubing Flexible Tubing	Group 15 & 49 MUST NOT EXCEED limitations specified under **Section 6–5–1** sub–heading **Shield/Cryo Cooler Compressor Water Cooling Lines & Hoses Requirements**
31	0.10 (66)	MSM1	RCP	50 (15.24) **minus takeup at both ends**	941	0.30 (7.62)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
49 SEE NOTE 1	—	Outdoor TSCC	Customer ball valves & hose barbs for Shield/Cryo Cooler	See Notes	— —	— —	copper supply line copper return line	copper supply line copper return line	• Customer to provide & install 0.75 (19 mm) ID copper lines for Shield/Cryo Cooler Compressor water cooling supply & return. • Customer to provide & install in Equipment Room in end of each copper line a 0.75 in. (19 mm) ball valve & 0.5 in. (12.7 mm) hose barb. • Groups 49 & 15 MUST NOT EXCEED limitations specified under **Section 6–5–1** sub–heading **Shield/Cryo Cooler Compressor Water Cooling Lines & Hoses Requirements**

Note 1 This Group contains water lines which shall be routed separate from electrical lines (i.e. power & signal).

(Continued)

6–5–1 Outdoor TSCC System Interconnects (Continued)

TABLE 6–4 (Continued)
OUTDOOR TSCC ADDITIONAL SYSTEM INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in.^{2}(mm^2)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO				FROM	TO	
50 SEE NOTE 1	—	Outdoor TSCC	Customer ball valves & hose barbs for Gradient Coil	See Notes	— —	— —	copper supply line copper return line	copper supply line copper return line	• Customer to provide & install 1.0 (25.4 mm) ID copper lines for Gradient Coil water cooling supply & return. • Customer to provide & install in Equipment Room in end of each copper line a 1 in. (25.4 mm) ball valve & 0.75 in. (19 mm) hose barb. • Group 50 and 86 MUST NOT EXCEED limitations specified under **Section 6–5–1** sub–heading **Gradient Coil Water Cooling Lines & Hoses Requirements**
51	1.097 (613)	Outdoor TSCC	RCP	See NOTES	— — —	0.63 (16.0) 0.65 (16.5) 0.76 (15.9)	1.125x2.00 (28.6x50.8) 1.125x2.00 (28.6x50.8) 1.50x2.25 (38.1x57.2)	1.125x2.00 (28.6x50.8) 1.125x2.00 (28.6x50.8) 1.50x2.25 (38.1x57.2)	Total length of 100 ft (30.5 m) minus height of connection at Outdoor TSCC and height of connection at RCP. No cable takeup inside of TSCC or RCP.
86 SEE NOTE 1	1.666 (1078)	Customer provided hose barbs	MG2	80 (24.4) minus takeup at hose barbs See Notes	536 537	1.03 (26.2) 1.03 (26.2)	Flexible Tubing Flexible Tubing	Flexible Tubing Flexible Tubing	• Group 86 and 50 MUST NOT EXCEED limitations specified under **Section 6–5–1** sub–heading **Gradient Coil Water Cooling Lines & Hoses Requirements** • These GEMS supplied Gradient Coil water cooling lines are routed through waveguides in PP1 and have a minimum bend radius of 7 in. (178 mm).
Note 1 This Group contains water lines which shall be routed separate from electrical lines (i.e. power & signal).									

6–5–2 Indoor Water Cooled TSCC System Interconnects

Note

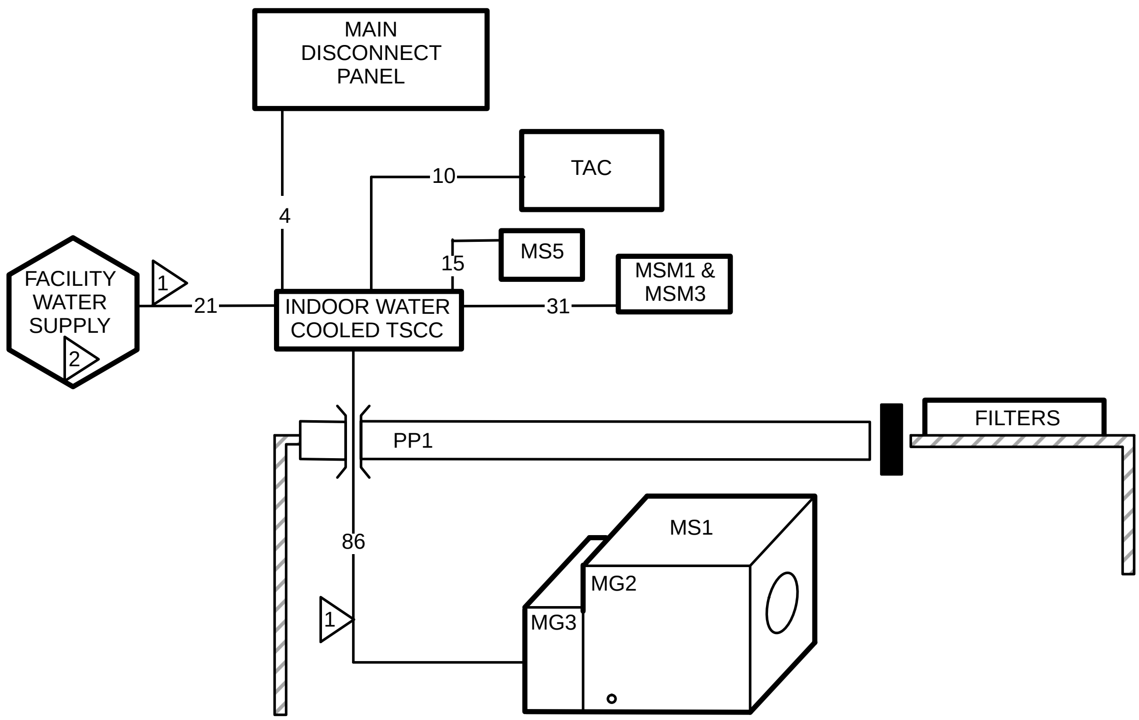
Refer to **Section 2–7–1 TSCC Configurations Siting Considerations** Table 2–5 listing of responsibility for the specific installation tasks. Also refer to *Ellis and Watts Technical Publication 471 Pre installation Installation Operating LC 18MT/WC Chilled Water System for GE Signa TwinSpeed MRI Equipment* for additional information and details.

Installation of the Indoor Water Cooled TSCC must be in accordance with local and national codes.

Illustration 6–4 shows the Indoor Water Cooled TSCC subsystem additional system Group Interconnect Diagram. Each group contains one or more cables, refer to Table 6–5 for specifics.

NOTE:

- ONLY INTERCONNECTS SPECIFIC TO TSCC SUBSYSTEM EQUIPMENT SHOWN HERE. REFER TO **Section 6–4 SYSTEM INTERCONNECTS** FOR OTHER SYSTEM INTERCONNECTS.



1 THIS GROUP CONTAINS WATER LINES WHICH SHALL BE ROUTED SEPARATE FROM ELECTRICAL LINES (I.E. POWER & SIGNAL).

2 FOR INDOOR WATER COOLED TSCC ONLY. REFER TO **Section 4–4–3 Water Cooled System Cooling Cabinet Configuration Requirements**

INDOOR WATER COOLED TSCC SUBSYSTEM GROUP INTERCONNECT DIAGRAM
ILLUSTRATION 6–4

6–5–2 Indoor Water Cooled TSCC (Continued)

TABLE 6–5
INDOOR WATER COOLED TSCC ADDITIONAL SYSTEM INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in.^{2}(mm^2)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO				FROM	TO	
4	—	MDP	TSCC	—	—	—	—	—	Power wiring Customer Furnished.
10	0.11 (71)	TSCC	TAC	50 (15.24)	992	0.375 (9.50)	1.60x2.00 (41.1x50.8)	1.60x2.00 (41.1x50.8)	Data cable
15 SEE NOTE 1	1.666 (1078)	TSCC	MS5	55 (16.7)	— —	0.75 (19.0) 0.75 (19.0)	rubber hose rubber hose	rubber hose rubber hose	See Notes 2 & 3 for vertical separation limits & TSCC hoses/connections.
21 SEE NOTE 1	—	Facility Water Supply	TSCC	—	—	—	—	—	For Indoor Water Cooled TSCC ONLY: Customer furnished, refer to **Section 4–4–3 Water Cooled System Cooling Cabinet Configuration Requirements** & refer to ***Note 5***
31	0.10 (66)	MSM1	TSCC	38 (11.6) **minus** takeup at MSM1	941	0.30 (7.62)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
86 SEE NOTE 1	1.666 (1078)	TSCC	MG2	80 (24.4)	536 537	1.03 (26.2) 1.03 (26.2)	Flexible Tubing Flexible Tubing	Flexible Tubing Flexible Tubing	These water cooling lines are routed through waveguides in PP1 and have a minimum bend radius of 7 in. (178 mm). See Notes 2 & 4 for vertical separation limits & TSCC hoses/connections.

Note 1 This Group contains water lines which shall be routed separate from electrical lines (i.e. power & signal).

2 The Indoor Water Cooled TSCC must not exceed 15 ft (4.6 m) vertical separation above or below the Shield/Cryo Cooler Compressor Cabinet **and** the Gradient Coil.

3 The Indoor Water Cooled TSCC has two 0.5 in. (12.7 mm) Snaptite OD brass nipples at the TSCC for the Shield/Cryo Cooler Compressor water cooling lines connections, refer to Illustration 2–17 for location of fittings on TSCC. Ellis & Watts provides the water cooling rubber hoses to connect from the Indoor Water Cooled TSCC to the Shield/Cryo Cooler Compressor.

4 The Indoor Water Cooled TSCC has two 0.75 in. (19 mm) Snaptite OD brass nipples at the TSCC for the Gradient Coil water cooling lines connections, refer to Illustration 2–17 for location of fittings on TSCC. GE Medical Systems provides two 0.75 inch (19 mm) ID rubber hoses of 100 ft (30.5 m) total length [usable length of 80 ft (24.4 m) minus takeups] to connect from TSCC to Gradient Coil inside the Magnet Enclosure.

5 Facility chilled water connections at TSCC are 1.5 inch (38.1 mm) FNPT. Customer to connect with 1.5 inch (38.1 mm) MNPT.

6–5–3 Indoor Air Cooled TSCC System Interconnects

Note

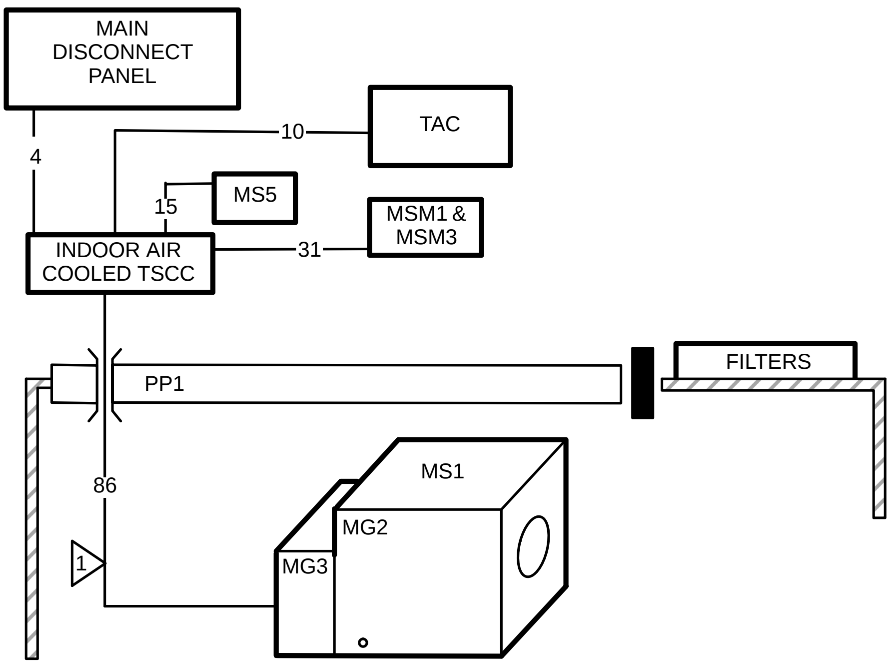
Refer to **Section 2–7–1 TSCC Configurations Siting Considerations** Table 2–5 for listing of responsibility for the specific installation tasks. Also refer to *Ellis and Watts Technical Publication 472 Pre installation Installation Operating LC 20MT Chilled Water System for GE Signa TwinSpeed MRI Equipment* for additional information and details.

Installation of the Indoor Air Cooled TSCC must be in accordance with local and national codes.

Illustration 6–4 shows the Indoor Air Cooled TSCC subsystem additional system Group Interconnect Diagram. Each group contains one or more cables, refer to Table 6–5 for specifics.

NOTE:

- ONLY INTERCONNECTS SPECIFIC TO TSCC SUBSYSTEM EQUIPMENT SHOWN HERE. REFER TO **Section 6–4 SYSTEM INTERCONNECTS** FOR OTHER SYSTEM INTERCONNECTS.



1 THIS GROUP CONTAINS WATER LINES WHICH SHALL BE ROUTED SEPARATE FROM ELECTRICAL LINES (I.E. POWER & SIGNAL).

INDOOR AIR COOLED TSCC SUBSYSTEM GROUP INTERCONNECT DIAGRAM
ILLUSTRATION 6–5

6–5–3 Indoor Air Cooled TSCC System Interconnects (Continued)

TABLE 6–6
INDOOR AIR COOLED TSCC ADDITIONAL SYSTEM INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in.^{2}(mm^2)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO				FROM	TO	
4	—	MDP	TSCC	—	—	—	—	—	Power wiring Customer Furnished.
10	0.11 (71)	TSCC	TAC	50 (15.24)	992	0.375 (9.50)	1.60x2.00 (41.1x50.8)	1.60x2.00 (41.1x50.8)	Data cable
15 SEE NOTE 1	1.666 (1078)	TSCC	MS5	55 (16.7)	— —	0.75 (19.0) 0.75 (19.0)	Flexible Tubing Flexible Tubing	Flexible Tubing Flexible Tubing	See Notes 2 & 3 for vertical separation limits & TSCC hoses/connections.
31	0.10 (66)	MSM1	TSCC	40 (12.2) **minus** takeup at MSM1	941	0.30 (7.62)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
86 SEE NOTE 1	1.666 (1078)	TSCC	MG2	80 (24.4)	536 537	1.03 (26.2) 1.03 (26.2)	Flexible Tubing Flexible Tubing	Flexible Tubing Flexible Tubing	These water cooling lines are routed through waveguides in PP1 and have a minimum bend radius of 7 in. (178 mm). See Notes 2 & 4 for vertical separation limits & TSCC hoses/connections.

Note 1 This Group contains water lines which shall be routed separate from electrical lines (i.e. power & signal).

2 The Indoor Air Cooled TSCC must not exceed 15 ft (4.6 m) vertical separation above or below the Shield/Cryo Cooler Compressor Cabinet **and** the Gradient Coil.

3 The Indoor Air Cooled TSCC has two 0.5 in. (12.7 mm) Snaptite OD brass nipples at the TSCC for the Shield/Cryo Cooler Compressor water cooling lines connections, refer to Illustration 2–19 for location of fittings on TSCC. Ellis & Watts provides the water cooling rubber hoses to connect from the Indoor Water Cooled TSCC to the Shield/Cryo Cooler Compressor.

4 The Indoor Air Cooled TSCC has two 0.75 in. (19 mm) Snaptite OD brass nipples at the TSCC for the Gradient Coil water cooling lines connections, refer to Illustration 2–19 for location of fittings on TSCC. GE Medical Systems provides two 0.75 inch (19 mm) ID rubber hoses of 100 ft (30.5 m) total length [usable length of 80 ft (24.4 m) minus takeups] to connect from TSCC to Gradient Coil inside the Magnet Enclosure.

6–6 TGWC ADDITIONAL INTERCONNECTS

The following sections show the system interconnects required for the Twin Gradient Water Cooler (TGWC) configurations in addition to the interconnects in **Section 6–4 SYSTEM INTERCONNECTS**.

6–6–1 Outdoor TGWC System Interconnects

Note

The TGWC provides water cooling for the Gradient Coil ONLY. Therefore customer provided water cooling is required for the water cooled Shield/Cryo Cooler Compressor.

Note

Refer to **Section 2–7–2 TGWC Configurations Siting Considerations** Tables 2–6 for listing of responsibility for the specific installation tasks. Also refer to *Technical Publication 474 Pre installation Installation Operating LC 10MT/O Chilled Water System for GE Signa TwinSpeed MRI Equipment* for additional information and details.

The site design for Outdoor TGWC installation must meet the following requirements for vertical separation and water lines/hose lengths limitations. **Installation of the Loop Outdoor TGWC must be in accordance with local and national codes.**

Vertical Separation Requirements

- Single Loop Outdoor TGWC must not exceed 100 ft (30.5 m) vertical separation from the Gradient Coil.

Gradient Coil Water Cooling Lines & Hoses Requirements

- Ellis & Watts provides two 1.0 in. (25.4 mm) FNPT fittings at Outdoor TGWC for the Gradient Coil water cooling lines connections, refer to Illustration 2–14 for location of fittings on TGWC. The Outdoor TGWC has ball valves located inside the cabinet. Customer to provide 1.0 (25.4 mm) MNPT connections and 1.0 in. (25.4 mm) ID copper lines.
- GE Medical Systems provides two 3/4 inch (19 mm) ID rubber hoses of 100 ft (30.5 m) total length [usable length of 80 ft (24.4 m) minus takeup for ball valve and hose barb location] to connect from Customer provided copper line hose barbs with ball valves to Gradient Coil inside the Magnet Enclosure. Site layout for the Gradient Coil water cooling lines and hoses must meet the following:
  - Total line length from the Outdoor TGWC to the Gradient Coil MUST NOT EXCEED 200 ft (61 m) from TGWC connection to Gradient Coil connection. GE supplied rubber hoses for Gradient Coil is a maximum length of 100 ft (30.5 m) total length [80 ft (24.4 m) usable length].
  - If the 1 in. (25.4 mm) copper lines exceed 100 ft (30.5 m) then the rubber hose length must be reduces 1 ft (0.3 m) for every 1 ft (0.3 m) of copper line that exceeds 100 ft (30.5 m).
  - Copper lines must be insulated outdoors.

RCP Data Cables Requirements

- Ellis & Watts provides two data cables that connect between the RCP and the Outdoor TGWC. These cables are 100 ft (30.5 m) total length. Usable length is dependent on TGWC and RCP placement [total length minus height of connection at Outdoor TGWC and height of connection at RCP, no cable takeup inside of TGWC or RCP].

Note

Contact Ellis & Watts International to determine if additional length cable are possible based on specific site design.

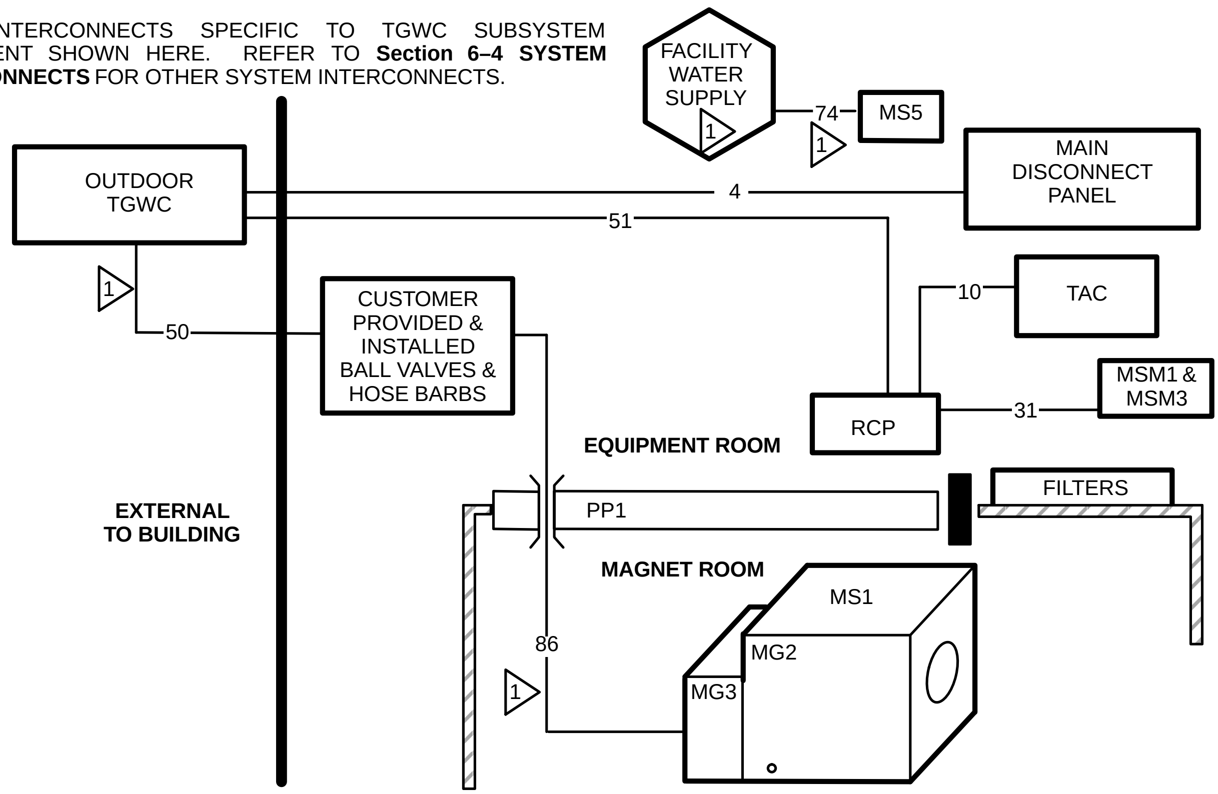
6–6–1 Outdoor TGWC System Interconnects (Continued)

System Additional Interconnects

Illustration 6–6 shows the Outdoor TGWC subsystem additional system Group Interconnect Diagram. Each group contains one or more cables, refer to Table 6–7 for specifics.

NOTE:

- ONLY INTERCONNECTS SPECIFIC TO TGWC SUBSYSTEM EQUIPMENT SHOWN HERE. REFER TO **Section 6–4 SYSTEM INTERCONNECTS** FOR OTHER SYSTEM INTERCONNECTS.



1 THIS GROUP CONTAINS WATER LINES WHICH SHALL BE ROUTED SEPARATE FROM ELECTRICAL LINES (I.E. POWER & SIGNAL).

OUTDOOR TGWC & RCP SUBSYSTEM GROUP INTERCONNECT DIAGRAM
ILLUSTRATION 6–6

6–6–1 Outdoor TGWC System Interconnects (Continued)

TABLE 6–7
OUTDOOR TGWC ADDITIONAL SYSTEM INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in.^{2}(mm^2)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO				FROM	TO	
4	—	MDP	Out-door TGWC	—	—	—	—	—	Power wiring Customer Furnished.
10	0.11 (71)	TGWC	TAC	50 (15.24)	992	0.375 (9.50)	1.60x2.00 (41.1x50.8)	1.60x2.00 (41.1x50.8)	Data cable
31	0.10 (66)	MSM1	RCP	60 (18.3) **minus takeup at both ends**	941	0.30 (7.62)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
50 SEE NOTE 1	—	Out-door TGWC	Cus-tomer ball valves & hose barbs for Gradi-ent Coil	See Notes	— —	— —	copper supply line copper return line	copper supply line copper return line	• Customer to provide & install 1.0 (25.4 mm) ID copper lines for Gradient Coil water cooling supply & return. • Customer to provide & install in Equipment Room in end of each copper line a 1 in. (25.4 mm) ball valve & 0.75 in. (19 mm) hose barb. • Group 50 and 86 MUST NOT EXCEED limitations specified under **Section 6–5–1** sub–heading **Gradient Coil Water Cooling Lines & Hoses Requirements**
51	4.791 (3091)	Out-door TGWC	RCP	See NOTES	— —	1.59 (40.4) 1.89 (48.0)	1.97x2.25 (50.0x57.2) 1.75x2.25 (44.5x57.2)	1.97x2.25 (50.0x57.2) 1.75x2.25 (44.5x57.2)	Total length of 100 ft (30.5 m) minus height of connection at Outdoor TGWC and height of connection at RCP. No cable takeup inside of TGWC or RCP.
74 SEE NOTE 1	—	Water	MS5	—	—	—	—	—	Customer Furnished water lines (refer **to Sections 4–4–4 and 5–1**).
86 SEE NOTE 1	1.666 (1078)	Cus-tomer pro-vided hose barbs	MG2	80 (24.4) minus takeup at hose barbs See Notes	536 537	1.03 (26.2) 1.03 (26.2)	Flexible Tubing Flexible Tubing	Flexible Tubing Flexible Tubing	• Group 86 and 50 MUST NOT EXCEED limitations specified under **Section 6–5–1** sub–heading **Gradient Coil Water Cooling Lines & Hoses Requirements** • These GEMS supplied Gradient Coil water cooling lines are routed through waveguides in PP1 and have a minimum bend radius of 7 in. (178 mm).

Note 1 This Group contains water lines which shall be routed separate from electrical lines (i.e. power & signal).

6–6–2 Indoor Water Cooled TGWC System Interconnects

Note

The TGWC provides water cooling for the Gradient Coil ONLY. Therefore customer provided water cooling is required for the water cooled Shield/Cryo Cooler Compressor.

Note

Refer to **Section 2–7–2 TGWC Configurations Siting Considerations** Tables 2–6 for listing of responsibility for the specific installation tasks. Also refer to *Technical Publication 469 Pre installation Installation Operating LTL – 10 Chilled Water System for GE Twin Magnet MRI Equipment* for additional information and details.

Installation of the Indoor Water Cooled TGWC must be in accordance with local and national codes.

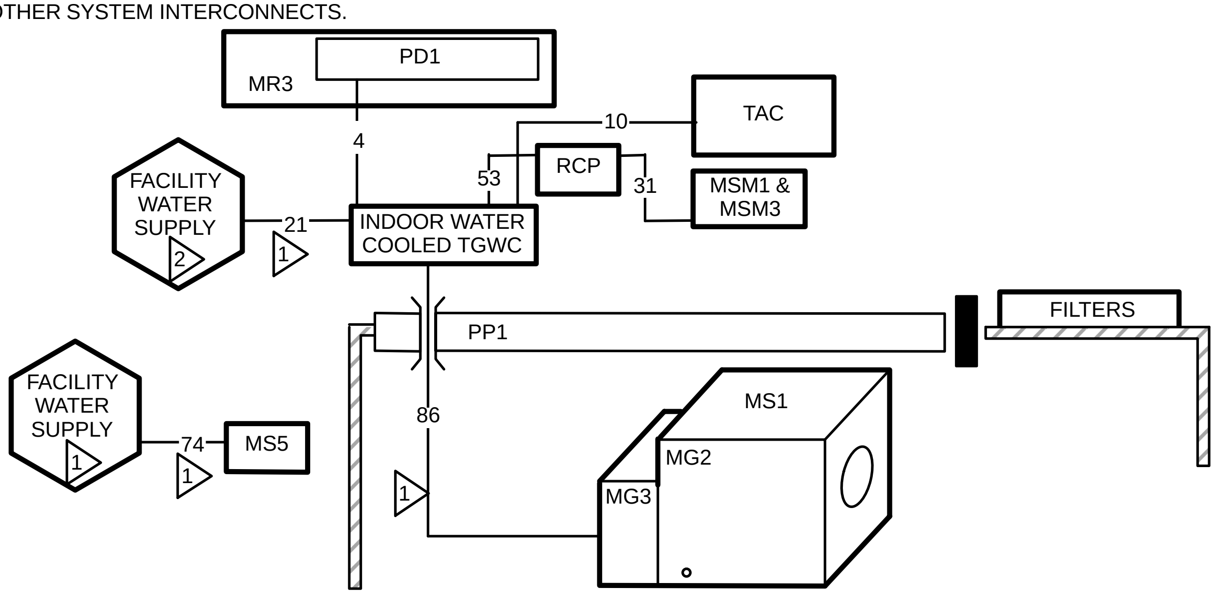
Illustration 6–7 shows the Indoor Water Cooled TGWC subsystem additional system Group Interconnect Diagram. Each group contains one or more cables, refer to Table 6–8 for specifics.

Note

The Indoor Water Cooled TGWC and Gradient Coil located inside the magnet must not be separated by a distance greater than 100 ft (30.5 m) of water lines. The TGWC vertical separation must not exceed 15 ft (4.6 m) above or below the Gradient Coil.

NOTE:

- ONLY INTERCONNECTS SPECIFIC TO TGWC SUBSYSTEM EQUIPMENT SHOWN HERE. REFER TO **Section 6–4 SYSTEM INTERCONNECTS** FOR OTHER SYSTEM INTERCONNECTS.



1. THIS GROUP CONTAINS WATER LINES WHICH SHALL BE ROUTED SEPARATE FROM ELECTRICAL LINES (I.E. POWER & SIGNAL).
2. FOR INDOOR WATER COOLED TGWC ONLY. REFER TO Section 4–4–3 **Water Cooled System Cooling Cabinet Configuration Requirements**

INDOOR WATER COOLED TGWC SUBSYSTEM GROUP INTERCONNECT DIAGRAM
ILLUSTRATION 6–7

6–6–2 Indoor Water Cooled TGWC (Continued)

TABLE 6–8
INDOOR WATER COOLED TGWC ADDITIONAL SYSTEM INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in.2(mm2)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO				FROM	TO	
4	0.159 (103)	PD1 **See Note 2**	TGWC	23 (7.0)	—	0.45 (11.43)	Hard Wired	1.53x2.78 (38.9x70.6)	Ellis & Watts provided cable
10	0.11 (71)	TGWC	TAC	50 (15.24)	992	0.375 (9.50)	1.60x2.00 (41.1x50.8)	1.60x2.00 (41.1x50.8)	Data cable
21 SEE NOTE 1	—	Facility Water Supply	TGWC	—	—	—	—	—	For Indoor Water Cooled TGWC ONLY: Customer furnished, refer to **Section 4–4–5 Water Cooled Twin Gradient Water Cooler (TGWC) Requirements**
31	0.10 (66)	MSM1	RCP	50 (15.24) **minus** takeup at booth ends	941	0.30 (7.62)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
53	0.157 (103)	Water Cooled TGWC	RCP	28 (8.53) **minus** takeup at RCP See Notes	— — —	0.258 (6.6) 0.258 (6.6) 0.258 (6.6)	1.00x1.48 (25.4x37.6) 1.125x2.0 (28.6x50.8) 1.125x2.0 (28.6x50.8)	1.00x1.48 (25.4x37.6) 1.125x2.0 (28.6x50.8) 1.125x2.0 (28.6x50.8)	No cable takeup inside of RCP.
74 SEE NOTE 1	—	Facility Water Supply	MS5	—	—	—	—	—	Customer Furnished water lines (refer to **Sections 4–4–4 and 5–1**).
86 SEE NOTE 1	1.666 (1078)	TGWC	MG2	80 (24.4)	536 537	1.03 (26.2) 1.03 (26.2)	Flexible Tubing Flexible Tubing	Flexible Tubing Flexible Tubing	These water cooling lines are routed through waveguides in PP1 and have a minimum bend radius of 7 in. (178 mm). See **Notes 3 & 4** for vertical separation limits & TGWC hoses/connections.

Note 1 This Group contains water lines which shall be routed separate from electrical lines (i.e. power & signal).

2 The PDU is a module (PD1) in the lower portion of the ACGD/PDU Cabinet (MR3).

3 Indoor Water Cooled TGWC must not exceed 15 ft (4.6 m) vertical separation above or below the Gradient Coil.

4 The Indoor Water Cooled TGWC has two 0.75 in. (19 mm) Snaptite OD brass nipples at the TGWC for the Gradient Coil water cooling lines connections, refer to Illustration 2–24 for location of fittings on TGWC. GE Medical Systems provides two 0.75 inch (19 mm) ID rubber hoses of 100 ft (30.5 m) total length [usable length of 80 ft (24.4 m) minus takeups] to connect from TGWC to Gradient Coil inside the Magnet Enclosure.

6–6–3 Indoor Air Cooled TGWC System Interconnects

Note

The TGWC provides water cooling for the Gradient Coil ONLY. Therefore customer provided water cooling is required for the water cooled Shield/Cryo Cooler Compressor.

Note

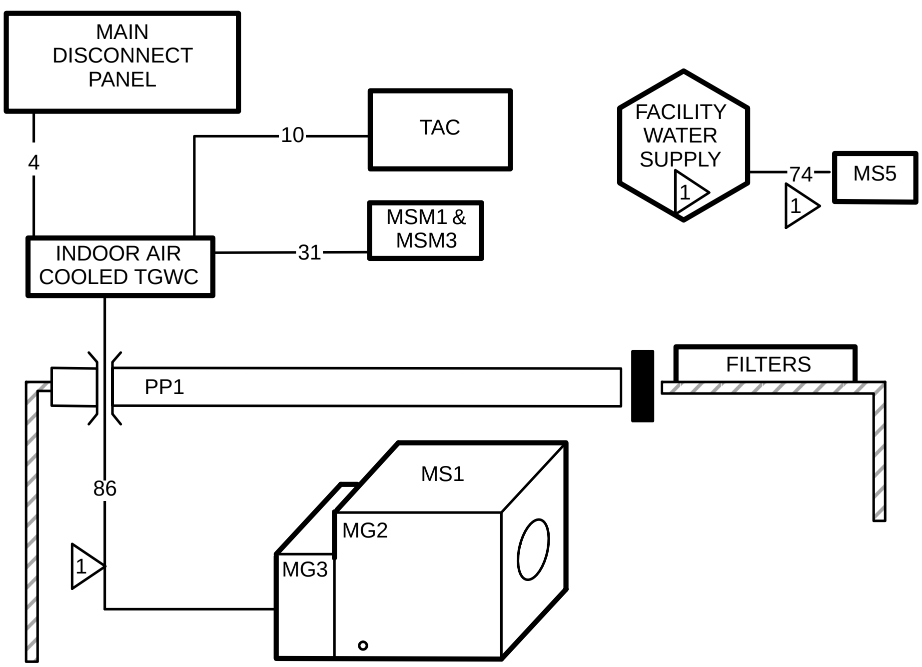
Refer to **Section 2–7–2 TGWC Configurations Siting Considerations** Tables 2–6 for listing of responsibility for the specific installation tasks. Also refer to *Technical Publication 473 Pre installation Installation Operating LC 10MT Chilled Water System for GE Signa TwinSpeed MRI Equipment* for additional information and details.

Installation of the Indoor Air Cooled TGWC must be in accordance with local and national codes.

Illustration 6–8 shows the Indoor Air Cooled TGWC subsystem additional system Group Interconnect Diagram. Each group contains one or more cables, refer to Table 6–9 for specifics.

NOTE:

- ONLY INTERCONNECTS SPECIFIC TO TSCC SUBSYSTEM EQUIPMENT SHOWN HERE. REFER TO **Section 6–4 SYSTEM INTERCONNECTS** FOR OTHER SYSTEM INTERCONNECTS.



1 THIS GROUP CONTAINS WATER LINES WHICH SHALL BE ROUTED SEPARATE FROM ELECTRICAL LINES (I.E. POWER & SIGNAL).

INDOOR AIR COOLED TGWC SUBSYSTEM GROUP INTERCONNECT DIAGRAM
ILLUSTRATION 6–8

6–6–3 Indoor Air Cooled TGWC (Continued)

TABLE 6–9
INDOOR AIR COOLED TGWC ADDITIONAL SYSTEM INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in.^{2}(mm^2)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO				FROM	TO	
4	—	MDP	TGWC	—	—	—	—	—	Customer Furnished.
10	0.11 (71)	TGWC	TAC	50 (15.24)	992	0.375 (9.50)	1.60x2.00 (41.1x50.8)	1.60x2.00 (41.1x50.8)	Data cable
31	0.16 (103)	MSM1	TGWC	48 (14.6) **minus** takeup at MSM1	941	0.30 (7.62)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
74 SEE NOTE 1	—	Facility Water Supply	MS5	—	—	—	—	—	Customer Furnished water lines (refer to **Sections 4–4–4 and 5–1**).
86 SEE NOTE 1	1.666 (1078)	TGWC	MG2	80 (24.4)	536 537	1.03 (26.2) 1.03 (26.2)	Flexible Tubing Flexible Tubing	Flexible Tubing Flexible Tubing	These water cooling lines are routed through waveguides in PP1 and have a minimum bend radius of 7 in. (178 mm). See Notes 2 & 3 for vertical separation limits & TGWC hoses/connections.

Note 1 This Group contains water lines which shall be routed separate from electrical lines (i.e. power & signal).

2 Indoor Air Cooled TGWC must not exceed 15 ft (4.6 m) vertical separation above or below the Gradient Coil.

3 The Indoor Air Cooled TGWC has two 0.75 in. (19 mm) Snaptite OD brass nipples at the TGWC for the Gradient Coil water cooling lines connections, refer to Illustration 2–26 for location of fittings on TGWC. GE Medical Systems provides two 0.75 inch (19 mm) ID rubber hoses of 100 ft (30.5 m) total length [usable length of 80 ft (24.4 m) minus takeups] to connect from TGWC to Gradient Coil inside the Magnet Enclosure.

6–7 CONTRACTOR FURNISHED COMPONENTS

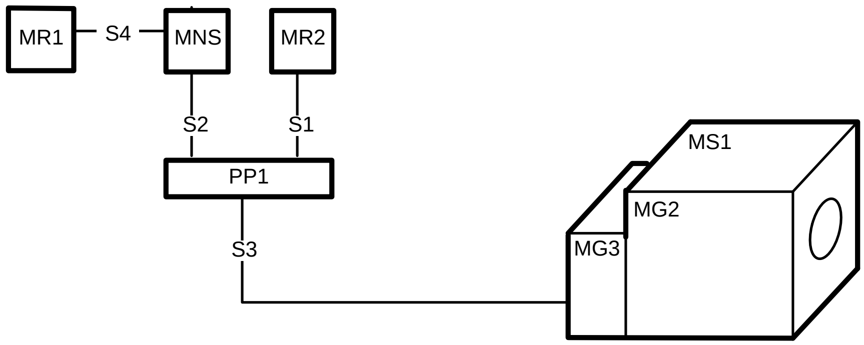
Table 6–10 lists contractor furnished components and details for connections to the system.

TABLE 6–10
CONTRACTOR FURNISHED COMPONENTS

ASSOCIATED EQUIPMENT	MATERIAL/LABOR PROVIDED BY CUSTOMER CONTRACTOR
POWER IN MAGNET ROOM	Provide and install power and wall duct for magnet rundown unit. (See **Section 5, POWER REQUIREMENT** for power specifications.)
SYSTEM GROUND	Provide ground cable between RF shielded room common ground point and Power Distribution Unit (PD1). (See **Section 5–4 GROUNDING** for cable specifications.)
EQUIPMENT POWER • Main Disconnect Panel (MDP) • Magnet Rundown Unit • MDP to Power Distribution Unit (PDU) • MDP to Twin System Cooling Cabinet (TSCC) or Twin Gradient Water Cooler (TGWC) • Service Outlet in Magnet Room • * Oxygen Monitor • * Advantage Workstation	Provide and install power, duct work, receptacle, and coverplate for each item listed. (See **Section 5 POWER REQUIREMENTS** for power specifications.)
PLUMBING	Provide and install all water cooling equipment, see **Section 4 SITE ENVIRONMENT**, for customer supplied components & requirements.
CRYOGENIC VENTING	Provide and install cryogenic vent system. (See **Section 4–9 CRYOGENIC VENTING**, for vent specifications.)
ROOM VENTILATION	Provide and install all room ventilation equipment (e.g. Magnet Room exhaust fan) **Section 4–8 ROOM VENTILATION** for room ventilation specifications.
PENETRATION PANEL MOUNTING HARDWARE	RF shielded room vendor to provide appropriate mounting hardware for GE supplied penetration panel. (See **Section 7 RF SHIELDED ROOM**.)
RF DOOR SWITCH AND CABLING	RF shielded room vendor to provide and install RF door switches on all RF shielded room doors. All switches must be wired in series. GE supplies a 100 ft (30.5 m) cable from System Cabinet which is terminated with 2 leads. These leads are connected to the set of switches. Switches must be in the open position when RF door is open but closed when door is closed. (See **Section 7 RF SHIELDED ROOM**.)
Note * Optional Equipment. ** The Pneumatic Patient Alert Control Box can be powered from an outlet on the Operator Workspace.	

6–8 SPECTROSCOPY INTERCONNECTS

Illustration 6–9 shows the 1.5T system group interconnects for the MNS Cabinet and Spectroscopy Modules added to the Penetration Panel and Magnet Enclosure. Table 6–11 contains the cable data for these interconnects.



SPECTROSCOPY SYSTEM INTERCONNECTS
ILLUSTRATION 6–9

TABLE 6–11
SPECTROSCOPY 1.5T SYSTEM INTERCONNECT LIST

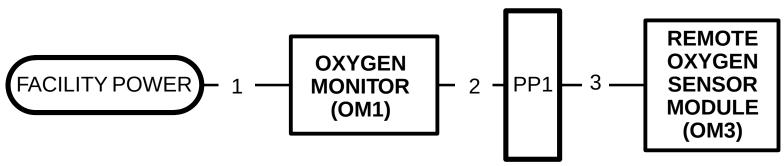
GROUP NUMBER	GROUP AREA in.^{2}(mm^2)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
S1	0.035 (22.8)	MR2	PP1	31 (9.45)	469	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	Additional interconnect to Table 6–3 Group 23.
S2	0.41 (267)	PP1	MNS	21 (6.40)	468	0.64 (16.3)	0.80x2.50 (20.3x63.5)	0.80x2.50 (20.3x63.5)	Additional interconnects to Table 6–3 Group 25.
S3	6.824 (4403)	PP1	MG3	30 (9.15)	472	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	Additional interconnects to Table 6–3 Group 45.
					472	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	Additional interconnects to Table 6–3 Group 45.
S4	0.656 (425)	MR1	MNS	9 (2.74)	936	0.68 (17.3)	2.00x4.00 (50.8x102)	2.00x3.63 (50.8x92.2)	Cabinet Power Cable, #10 AWG 3 wires
					961	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
					962	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					963	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					964	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					965	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	

6–9 OXYGEN MONITOR OPTION INTERCONNECTS

The Oxygen Monitor option consists of the following items:

- Oxygen Monitor
- Remote Oxygen Sensor Module
- Interconnect cables

Illustration 6–10 shows the Interconnect Diagram. Table 6–12 contains the cable data.



DISTANT REMOTE CONSOLE CABLING
ILLUSTRATION 6–10

TABLE 6–12
OXYGEN MONITOR INTERCONNECT LIST

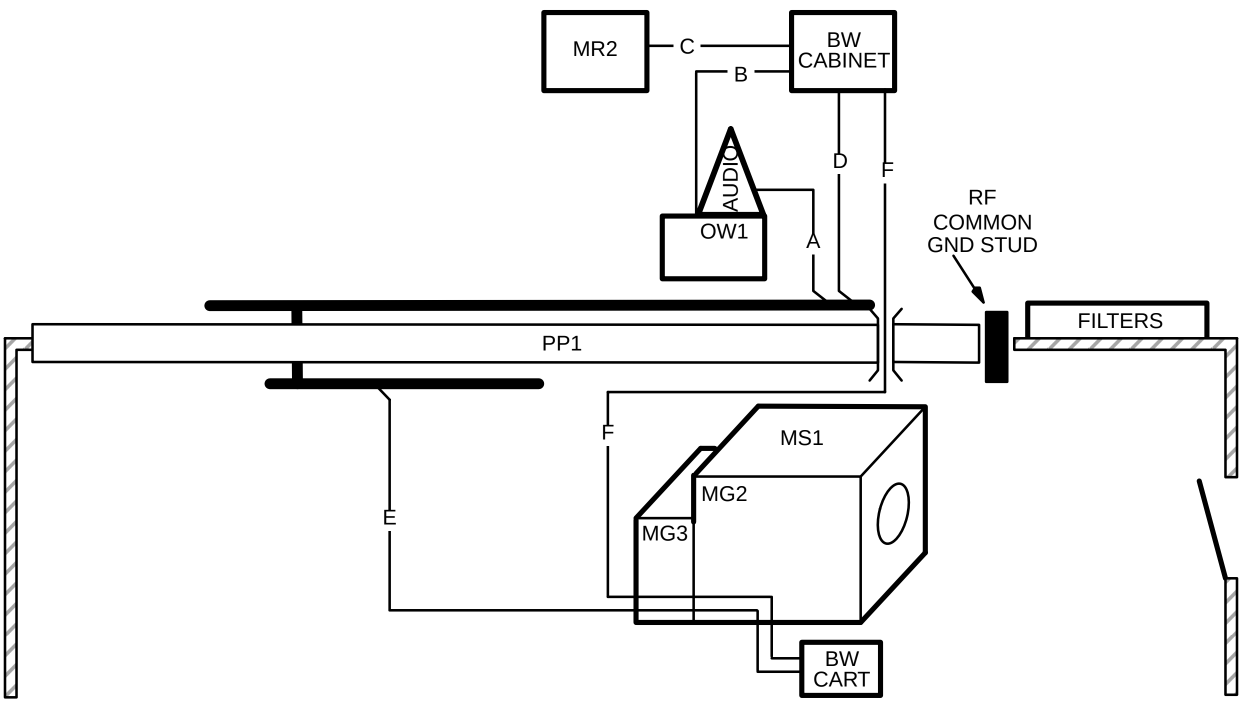
GROUP NUMBER	GROUP AREA in.^{2}(mm^2)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO				FROM	TO	
1	—	Facility Power	OM1	—	—	—	—	—	Customer Furnished recommended power source for OM1 (refer to **Section 5–1**).
2	0.096 (61.94)	OM1	PP1	94 (28.7) **minus** takeup at OM1	457	0.35 (8.9)	Hard Wired	1.30x2.00 (33.5x50.8)	
3	0.096 (61.94)	PP1	OM3	84 (25.6) **minus** takeup at OM3	458	0.35 (8.9)	1.30x2.00 (33.5x50.8)	Hard Wired	

6–10 BRAINWAVE OPTION INTERCONNECTS

The BrainWave option consists of the following items:

- BrainWave Audio Console
- BrainWave Cabinet
- BrainWave Cart
- Interconnect cables

Illustration 6–11 shows the Interconnect Diagram. Table 6–13 contains the cable data.



BRAINWAVE OPTION GROUP INTERCONNECT DIAGRAM
ILLUSTRATION 6–11

6–10 BRAINWAVE OPTION INTERCONNECTS (Continued)

TABLE 6–13
BRAINWAVE OPTION INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in.^{2}(mm^2)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER (Vendor number)	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO				FROM	TO	
A	0.113 (74)	Audio on OW1	PP1	92 (28.0)	758620	0.38 (9.7)	0.60x1.90 (15.2x48.3)	1.26x1.75 (32.0x44.5)	Audio power / signal cable
B	0.081 (53)	OW1	BW Cabinet	66 (20.1)	560213	0.20 (5.1)	1.66x0.38 (42.2x9.7)	1.55x0.95 (39.4x24.1)	560213 connects to Audio Console located on OW1
					758616	0.25 (6.4)	0.50x0.88 (12.7x22.3)	0.50x0.88 (12.7x22.3)	Network Cable
C	0.194 (125)	MR2	BW Cabinet	10.0 (3.0)	520423	0.38 (9.7)	1.45x2.5 (36.8x63.5)	1.45x2.5 (36.8x63.5)	Trigger Extension Cable
					560202	0.32 (8.1)	1.28x3.0 (32.5x76.2)	Hardwired	AC Power
D	0.227 (148)	BW Cabinet	PP1	60 (18.3)	979153	0.38 (9.7)	0.72x2.43 (18.3x61.7)	1.26x1.75 (32.0x44.5)	DC Power – response
					760065	0.38 (9.7)	0.6x1.9 (15.2x48.3)	2.2x2.5 (55.9x63.5)	DC power – video
E	0.340 (222)	PP1	BW Cart routed through MG2/3	53 (16.2) 53 (16.2)	770065	0.38 (9.7)	2.2x2.5 (55.9x63.5)	0.75x2.0 (19.1x50.8)	DC Power – vidoe. This run is routed through MG2/3 and out to the BW Cart.
					758619	0.38 (9.7)	1.26x1.75 (32.0x44.5)	0.60x1.90 (15.2x48.3)	Audio power/signal cabler. This run is routed through MG2/3 and out to the BW Cart.
					979152	0.38 (9.7)	1.26x1.75 (32.0x44.5)	0.72x2.43 (18.3x61.7)	DC power – Response This run contains a connector at MG2 with a separate cable routed out to the BW Cart.
F	0.239 (156)	BW Cabinet	BW Cart	106 (32.3)	758615	0.40 (10.2)	1.75x2.5 (44.5x63.5)	1.75x2.5 (44.5x63.5)	Video fiber optic These interconnects are routed through PP1 Waveguide, through MG3, & partially through MG2 (exits at midpoint of Enclosure side). Usable length allows 6 ft (1.8 m) from MG2 to BW Cart.
					781100	0.38 (9.7)	air tubing	air tubing	